Lecture 1: Course Overview & Introduction

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14.461, Fall 2026

Course logistics & overview

- **Instructor**: Christian Wolf
  - Lectures: Mon/Wed 10:30am-12:00pm (Room E52-432)
  - Office hours: by appointment (Room E52-520)
- **TA**: Jan Morgan
  - Recitations: Fri 2:30-4:00pm (Room E51-390)
  - Office hours: by appointment
- **Course website**: `https://canvas.mit.edu/courses/39004`
- **Topic**: identification in (business-cycle) macroeconomics

  How can we use macro & micro data to answer the standard business-cycle macro questions?

Readings

- The class will be largely organized around readings of **(recent) papers**
  - List of readings is provided on the syllabus
  - Starred (*) papers are required reading before class, rest provides more background. Please read the starred papers critically, thinking about key insights & central assumptions.
- There will be some review of **classical time series material**. For this I'd recommend consulting a textbook—my favorites include:
  - Brockwell & Davis (theory of covariance-stationary time series, not aimed at economists)
  - Hamilton (classic, slightly dated reference on time series econometrics)
  - Cochrane (well-written introduction to linear time series methods, for economists)
  - Kilian & Lütkepohl (recent book on SVAR methods)

Requirements, grading, and prerequisites

- **Requirements & grading**
  1. Class participation (30%)
  2. Problem set (30%) [mix of coding & analytical exercises]
  3. Referee report & paper replication (40%)
     - Select an unpublished paper that is (at least broadly) related to the course content, check with me first for approval, use it for both report and replication
     - Referee report: reference for refereeing best-practice is posted on the course webpage
     - Paper replication: reproduce main results, then investigate one (minor) extension
- **Course prerequisites**: 1st-year PhD course material (macro & econometrics)

Outline

1. Impulse-Propagation Paradigm
   - A Brief History of Thought
   - Using the SVMA
   - The Role of Policy

2. A Bird's-Eye View of the Course

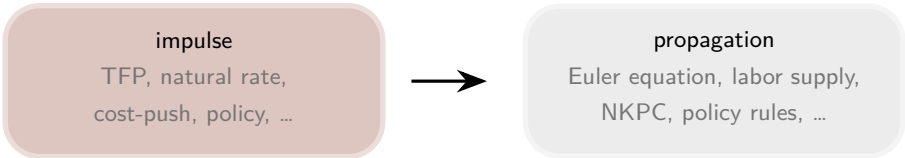
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Impulse-propagation paradigm

- Modern business-cycle macroeconomics adopts the "**impulse-propagation paradigm**"



  - **Business cycle** = endogenous response of the economy to random "shocks"
    Dominant methodological tradition since the 1960s. No (strong) endogenous boom-bust.
  - Criterion for success: match actual historical time series data. Two main paradigms:
    1. *Unconditional*: match average volatility of macro aggregates [recall RBC literature]
    2. *Conditional*: match specific historical episodes, incl. responses to specific "shocks"

- Next slide: mini **NK model** as running example of the impulse-propagation framework

Running example: NK model

- Recall the textbook **three-equation NK model** you saw in first year:

$$x_t = \mathbb{E}_t[x_{t+1}] - \frac{1}{\gamma}(i_t - \mathbb{E}_t[\pi_{t+1}]) + \varepsilon_t^d \quad \text{(IS)}$$

$$\pi_t = \kappa x_t + \beta \mathbb{E}_t[\pi_{t+1}] + \varepsilon_t^s \quad \text{(NKPC)}$$

$$i_t = \phi_\pi \pi_t + \varepsilon_t^m \quad \text{(TR)}$$

  - **Impulse**: shocks $\varepsilon_t^d, \varepsilon_t^s, \varepsilon_t^m$ to demand, supply, and monetary policy
  - **Propagation**: two private-sector relations [from optimizing behavior] + asn's on policy

- Assuming iid shocks, you can solve for a **static mapping** from shocks to outcomes:

$$\begin{pmatrix} x_t \\ \pi_t \\ i_t \end{pmatrix} = \underbrace{\frac{1}{\gamma + \kappa\phi_\pi} \begin{pmatrix} \gamma & -\phi_\pi & -1 \\ \gamma\kappa & \gamma & -\kappa \\ \gamma\kappa\phi_\pi & \gamma\phi_\pi & \gamma \end{pmatrix}}_{\equiv \Theta} \begin{pmatrix} \varepsilon_t^d \\ \varepsilon_t^s \\ \varepsilon_t^m \end{pmatrix}$$

Vector moving average representations

- The solution on the previous slide is in the form of a **vector moving average** model

$$\underbrace{y_t}_{\text{outcomes}} = \underbrace{\Theta}_{\text{propagation}} \times \underbrace{\varepsilon_t}_{\text{impulse}}$$

This particular VMA was simplified in two key respects:

1. *Static*: a dynamic linear structural VMA takes the form

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}$$

2. *Linear*: the most general impulse-propagation mapping takes the form

$$y_t = g(\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$$

- Why are VMAs the "right" way to think about **modeling the business cycle**?

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Two paths to business cycles

- Modeling of business cycles comes in **two broad flavors**:
  1. Business cycles result from forces *internal* to the economy
     - → Recurrent booms and busts, arising from eq'm incentives in decentralized economies
     - → Implication: get business cycles even absent any outside disturbances
  2. Business cycles result from forces *external* to the economy – **impulse-propagation**
     - → If no shocks ($\varepsilon_t = 0$), then the economy is stable
- A nice summary of the state of the literature is provided in Werning (2016, NBER MA):

> *Early theorizing on business cycles sought to generate **self-sustaining fluctuations**. However, Slutzky (1937) showed the way down a different path with linear systems continuously **buffered by shocks**. Subsequent modeling, the Wold decomposition and its empirical implementations, elevated these ideas to a dominant methodological tradition. Most of the macroeconomic canon today simply assumes business cycles are due to shocks.*

Why impulse-propagation?

- As said, modern macro follows the **impulse-propagation paradigm**
- Why? [only sketch here, can chat more if interested]
  1. *Model-based evidence*
     - → Desire to fit *unconditional* time series data tends to favor models where persistent exogenous driving forces are a key source of fluctuations [e.g., Kiyotaki-Moore (1997)]
     - → Those same models can fit the response of the economy to observed outside impulses
       E.g., sudden oil price increases, build-ups in military expenditure, ...
  2. *Direct empirical evidence*
     - → Internal vs. external cycles leave a different imprint on what's known as the spectral density of the data. Spectral densities of macro series look more consistent with external cycles.
     - → Recently debated, however! See Beaudry-Galizia-Portier (2020).
- Often – though not always – focus on *first-order* approximations = **linear SVMA(∞)**

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Using the SVMA

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}, \quad \varepsilon_t \sim N(0, I)$$

A key task for us will be to get the Θ's from a mix of models & data. To what end?

Using the SVMA

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1. Θ's = **dynamic causal effects**

$$\text{IRF}_{i,j,h} = \Theta_{i,j,h} \equiv \mathbb{E}(y_{i,t+h} \mid \varepsilon_{j,t} = 1) - \mathbb{E}(y_{i,t+h} \mid \varepsilon_{j,t} = 0), \quad h = 0, 1, 2, \ldots$$

   - Do oil shocks increase inflation? Are tariffs inflationary? The associated Θ's tell us.
   - Much of our focus will be on policy:
     - → Some Θ's are responses to policy *shocks*, e.g., military build-ups
     - → Systematic policy – like automatic stabilizers, or rate cuts in recessions – instead shapes *all* the Θ's. Lucas (1976) critique. Distinction will loom large throughout the class.

Using the SVMA

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2. Contribution of shocks to particular **historical episodes**

$$\text{HD}_{i,j,t} = \sum_{\ell=0}^{\infty} \Theta_{i,j,\ell} \varepsilon_{j,t-\ell} \equiv y^j_{i,t}$$

- The derived path $y^j_{i,t}$ is then compared with the actual realized time path $y_{i,t}$
- In words: did shock ε_j contribute meaningfully to a given historical episode?
  For example: How important were supply chain disruptions for the post-2021 inflation? Or AI for the recent investment boom?

Using the SVMA

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}, \quad \varepsilon_t \sim N(0, I)$$

A key task for us will be to get the Θ's from a mix of models & data. To what end?

3. **Shock importance** for average cyclical fluctuations

$$\text{FVD}_{i,j,h} \equiv 1 - \frac{\text{Var}(y_{i,t+h} \mid \{\varepsilon_{t-\ell}\}_{\ell=0}^{\infty}, \{\varepsilon_{j,t+\ell}\}_{\ell=1}^{h})}{\text{Var}(y_{i,t+h} \mid \{\varepsilon_{t-\ell}\}_{\ell=0}^{\infty})} = \frac{\sum_{m=0}^{h-1} \Theta_{i,j,m}^2}{\sum_{k=1}^{n_\varepsilon} \sum_{m=0}^{h-1} \Theta_{i,k,m}^2}$$

- In words: what share of fluctuations of y_{t+h} comes from shocks ε_j (rather than ε_{-j})?
- Speaks to the average cycle. Is the average contraction driven by "demand" or "supply"?

Decomposing business cycles

- Return to the three-equation NK model:

$$x_t = \mathbb{E}_t[x_{t+1}] - \frac{1}{\gamma}(i_t - \mathbb{E}_t[\pi_{t+1}]) + \varepsilon_t^d \quad \text{(IS)}$$

$$\pi_t = \kappa x_t + \beta \mathbb{E}_t[\pi_{t+1}] + \varepsilon_t^s \quad \text{(NKPC)}$$

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  - Given the ε_t's and Θ, we can decompose observed $\{x_t, \pi_t, i_t\}$ into shock contributions

- Through this lens, what is the "**contribution**" of policy to a given episode?
  - Causality is only assigned to the policy shock ε_t^m. Policy rule ϕ_π is part of "propagation".
  - Not in line with public commentary: if low rates post-2021 were part of the systematic rule (because ϕ_π was small), then do we say that policy didn't contribute to the inflation spike?

Intuitively, more natural to phrase causality in terms of the policy instrument, not just shocks ...

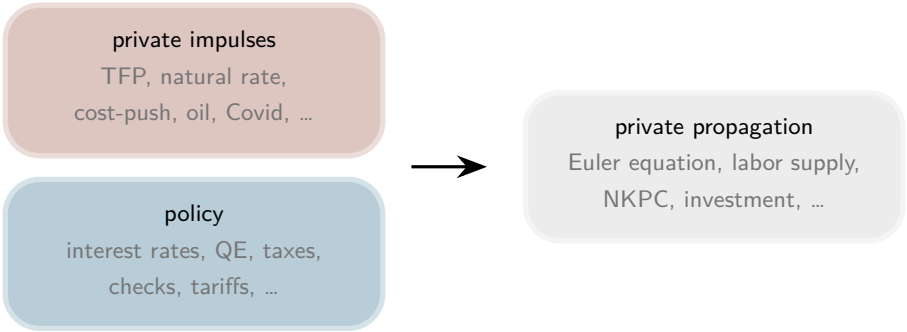
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A slightly revised impulse-propagation paradigm

- What if instead we viewed **policy** – not just policy shocks – as part of the **impulse**?



  - Here all of policy, both systematic & shocks, is part of an impulse to the private sector

- Now could change policy **as a whole** – rule & shocks – and ask what happens

  Are the causal effects of changes in the rule and changes in shocks fundamentally different?

Policy causal effects

- In the simple static NK model, **shocks and rules** are intimately connected

  1. *Shocks*

$$x_t = -\frac{1}{\gamma + \phi_\pi \kappa}\varepsilon_t^m, \quad \pi_t = -\frac{\kappa}{\gamma + \phi_\pi \kappa}\varepsilon_t^m, \quad i_t = \frac{\gamma}{\gamma + \phi_\pi \kappa}\varepsilon_t^m$$

     → Shock causal effects go through the policy rule as part of the propagation loop, so the effects depend on ϕ_π [Lucas critique: policy shapes eq'm relations]

  2. *Rule*: policy enters private sector through i_t, nothing else

$$x_t = -\frac{1}{\gamma} i_t, \quad \pi_t = -\frac{\kappa}{\gamma} i_t$$

     → Note how this is just the ratio of shock causal effects. Policy just matters through i_t.

- So here **shocks & rules** are two sides of the same coin – all through i_t. Is this general?

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class will be organized around the **(revised) impulse-propagation framework**

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1. Background material (lectures 2-3)

   a) Linearized **structural macro models**

      - Objective: compute the Θ's in structural equilibrium models
      - Methods for solving *linearized* business-cycle models: the state-space representation and the sequence-space representation

   b) Linear **time series methods**

      - Review classical topics like lag operators, linear filters, VARMA models, Wold decomposition theorem, spectral analysis
      - Note: necessary for empirical time series approaches to learn about the Θ's

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2. Policy and policy causality (lectures 4-6)
   - **History of thought**: Cowles Commission, Lucas critique and the rational expectations revolution, the Sims (1980) approach to empirical macroeconomics
   - Connection between **policy shocks** & systematic **policy rules**
     - Lucas-Sims bridge: policy – both systematic and shocks – acts through common, well-defined notion of "policy causal effects" = effects of the policy instrument
       Important subtlety: policy also plays an equilibrium selection role.
     - Rules and shocks are thus the same, and policy causal effects allow us to assess the contribution of systematic policy to any given historical episode
   - Formalizes the revised impulse-propagation paradigm: **shocks** + **policy** as impulse

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class will be organized around the **(revised) impulse-propagation framework**

3. Time series methods for shock & policy causal effect estimation (lectures 7-8)

   a) **Identification**

      - Challenge: from aggregate time series alone, SVMA models are (severely) under-identified
      - Classes of identifying asn's: invertibility + X, invertibility-robust methods (e.g., macro IVs)
        Overarching objective: talk about identification closer to credibility revolution language.

   b) **Estimation**

      - Review standard estimation methods (VARs, LPs) & intermediate shrinkage techniques (e.g., penalized LP), discuss best-practice for empirical analysis

   c) **Non-linear time series methods**
      Potential outcomes framework, LATE interpretation, ...

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4. Applications (lectures 9-10)
   a) **"What-if" scenarios for policy**
      - Would different interest rate policies have helped with inflation post-2021?
   b) **Origins of business cycles**
      - Are different business-cycle episodes caused by common types of shocks? Or is every recession different?
   c) **Estimating macroeconomic equations**
      - How to estimate macro equations like NKPCs? shocks as IVs vs. standard single-equation estimation methods

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5. Cross-sectional methods for shock & policy causal effect estimation (lectures 11-12)

   a) What does **cross-sectional analysis** actually identify?
   Key tool: PE-GE decompositions using sequence-space characterization of equilibria.

   b) How can we **aggregate** to recover macro shock effects?

      - Discuss two popular approaches to go from micro to macro: macro as explicitly aggregated micro and exploiting "shock commonality" in GE
      - Main application: stimulus check policies

   c) Can we use micro data to learn about the **channels** through which shocks & policies transmit in the macro-economy?